The Semantic Uncertainty Principle

The Physics of Language and Why Your Words Have a Margin of Error

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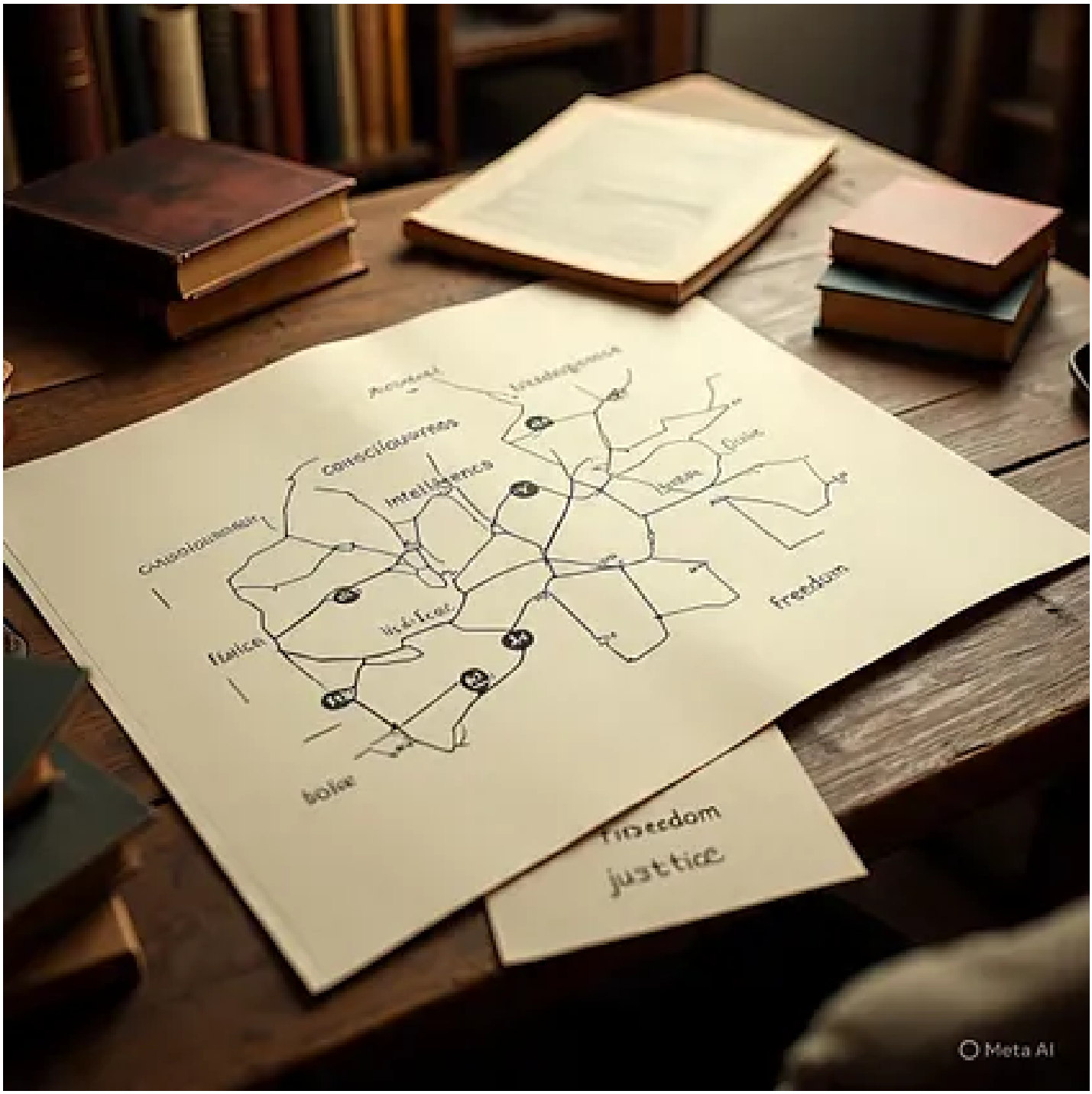
This article is a new, synthesis of ideas first explored in two academic papers from my Attralucian Essays series: "Semantic Uncertainty" and "Transfictors: Words as Compressed Transducers of Meaning."

In science, we are obsessed with uncertainty. A physicist measuring a particle, a biologist sequencing a gene, or an engineer testing a material would never present a result as a single, perfect number. They give us a value and with it a margin of error: 2.21 Volts, *plus or minus 0.01*. This margin of error isn't a sign of failure; it's a fundamental mark of intellectual honesty. It tells us the boundaries of our knowledge and what we can infer. The error comes from many places, the equipment, the recording process, even the person making the measurement. And there is always an error no measurement is ‘perfect’.



Yet in our most important tool, the language we use, we fail to do this every single day. In practice and often without thought or deeper consideration, We build entire conceptual cathedrals on words like "consciousness," "intelligence," "freedom," or "justice" as if they were solid a solid bedrock. We then go on to debate their meaning, we build policies around them, and we train our most advanced AIs on them. But then importantly, we rarely, if ever, acknowledge that these terms are not fixed symbols of certitude. As we lean into each symbol and try and uncover what it ‘means’ they become semantic clouds, shifting and unstable, the meaning shift in context and with each reader and reading. Any failure to account for this

instability becomes a sources of miscommunication and contest in science and society. This begs an important question!

What if we could treat words with the same rigor as numbers? What if we could develop a "physics of language" that accounts for their inherent uncertainty?

The Transducer Model: Your Words are Not Symbols, They're Sensors

To begin, we may need to update our model of what a word is. It’s commonplace to think of words as static labels that point to things in the world. But this is far too simple model for what is a very complex ‘signal’ of communication.

For me, a more accurate model, which I call the **Transducer Model**, frames words as finite, operational sensors. In this model just as a microphone transduces complex sound waves into a compressed electrical signal, a word transduces a messy, high-dimensional reality into a compressed linguistic signal.

- The word **"fire"** compresses a learned manifold of heat, color, hazard, emotion, and causality.
- The word **"consciousness"** compresses first-person awareness, functional cognition, philosophical dualisms, and cultural intuitions.
- The word **"chair"** compresses a vast, heterogeneous class of objects designed for sitting.
- Consider how the word **"red"** can mean a wavelength between two limits - this is just like a numbered measurement with an error i.e. 650nm plus or minus 50nm.

This act of compression is the ‘engine of linguistic efficiency’. It's a a wonderful and important property that allows us to communicate complex ideas without exhaustive descriptions. However, this compression comes at a cost is simplifies the signal. In effect It is finite, and what computer scientists call - lossy. The word is context sensitive, that is it can change signal depending on the words around it adding. And so It introduces a finite and real margin of error.

Fiction Quality: A Measurement for a Word's Precision

Let’s consider this further: If words are sensors, then their "readings" have a measurable precision. I call this a word's **"fiction quality" i.e.** a metric for its semantic stability, ranging from near 0% to 100%. This quality is directly and inversely related to its level of compression.

Low-Compression / High-Quality Words: A word like **"cheetah"** is a high-fidelity sensor. It refers to a specific, biologically defined entity. While not perfect, its meaning is highly stable across contexts. It requires very little compression because its referent is so well-defined. Its fiction quality is extremely high, perhaps **99%**.

High-Compression / Low-Quality Words: A word like **"freedom"** is a low-fidelity sensor. It compresses a monumental and abstract philosophical concept. Its meaning is almost entirely dependent on qualifiers ("freedom *of speech*," "financial *freedom*"). Without them, the word is a coarse reading with a massive margin of error. Its fiction quality might be as low as **20%**.

Now consider our humble **"chair."** This is a moderately compressed term. It simplifies a category of objects, but its meaning is highly ambiguous without more information. A request to "bring a chair" is a sensor reading of "medium," requiring follow-up measurements. Is it an office chair? A dining chair? A stool? Its fiction quality hovers around **50%**.

The Reader as Co-Author: Closing the Circuit

This brings us to the most crucial part of the model presented here: you, the reader. A reader is not a passive receiver of a signal. They are an active co-author of meaning. This is an idea that is widely understood in literature and philosophy but less so in science.

When you encounter a low-quality word like "chair," your brain immediately begins to decompress it by adding context-specific qualifiers. If you're in a conference room, you filter the signal to mean "ergonomic office chair." The reader's context adds their own unique and finer precision to the coarse signal sent by the writer based on their own body of knowledge and education.

This collaborative act of decompression is what makes language work. It's a complex dance between the writer's compression and the reader's decompression. The very real problem is that we rarely make the rules of this dance explicit. To refer to this idea I call it '**The Semantic Uncertainty Principle**'. So now we have this principle how can we use it?

The Semantic Uncertainty Appendix (SUA): A Proposal for Transparency

If the problem is hidden uncertainty, the solution may be to bring this uncertainty to the surface so that we have more transparency to help us with our thinking. This is why maybe we can

consider ideas such as a formal adoption of a Semantic Uncertainty Appendix (SUA) for any research, analysis, or article that relies on abstract or language-dependent terms.

The SUA is maybe not perfect and at the moment just a speculative idea. It could, for example, constitute a simple, structured system for disclosing the semantic scaffolding upon which a works ideas rest. Especially dense academic works that are given authority. In this way it’s a practical tool for making the "fiction quality" of our words measurable. It would require an author to outline:

1. **Operational Definitions:** How, exactly, are you using this term for this specific context?
2. **Known Ambiguities:** What are the common misinterpretations or alternative meanings you are intentionally excluding?
3. **Domains of Validity:** In what fields or contexts is this definition applicable? Where does it start to drift?
4. **Justification:** Why was this term chosen over another (e.g., "intelligence" vs. "understanding")?

An SUA for "cheetah" would be short and simple. An SUA for "consciousness" would be long, complex, and profoundly revealing. Such an approach would ask the author to be honest about the boundaries of their claim, and it would empower the reader to understand consider their own definitions within both their own context and the work presented.

Why This Matters Now More Than Ever

In an age where artificial intelligence is trained on the entirety of human language, maybe semantic accountability should not optional. Think about the implications and impact words have on peoples everyday lives. An AI model's "understanding" of a word is a reflection of it’s learned regularities of our own compressed, ambiguous language. In this light, understanding fiction quality could be crucial for preventing model collapse and improving the interpretability and safety of AI systems.

Furthermore, this framework may provide a diagnostic tool for improving public discourse. Consider that by identifying and qualifying high-compression terms such as "sustainability," "democracy," or "harm" we can reduce the friction of misinterpretation and foster a shared understanding built on the core idea that words are imperfect and that we as readers give them value and meaning.

It seems to me, that to treat our words as fixed, perfect instruments is to build our conceptual world on unstable ground. As we enter into a possible new age of enlightenment where AI and LLMs ask to think more about words and meaning It's time we considered acknowledging their margins of error. It's possibly time we adopted a ‘physics’ of language and take to heart the *Semantic Uncertainty Principle*.

As we rely more on AI and LLMs, the bigger questions may be about how we solve this 'semantic uncertainty' problem and what are the biggest risks if we don't?

How does semantic uncertainty effect your own reflections on what you read?

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